

IAL Physics Unit 2 — Resistivity & Core Practical 7 | Recall Pack

IAL Physics Unit 2 — Resistivity & Core Practical 7

The whole topic in one mental model. Resistance $R = V/I$ is a property of a *component* — change its length or thickness and R changes. Resistivity ρ is the deeper, intrinsic property of the *material* itself: it strips out the geometry. The bridge between them is $R = \rho l/A$ — a longer wire has more R ($R \propto l$), a fatter wire has less ($R \propto 1/A$), and the constant of proportionality is ρ . Core Practical 7 is just this equation turned into a straight-line graph: plot R (y) against l (x), the gradient is ρ/A , so $\rho = \text{gradient} \times A$, where the area $A = \pi d^2/4$ comes from a micrometer reading of the diameter (in **mm**, converted with $\times 10^{-3}$). Two recall anchors rule everything: $R = V/I$ **never** $R = VI$; and the unit of ρ is $\Omega \text{ m}$ **never** Ω/m .

1. Jargon glossary

Term	Student-friendly meaning	Exact physics meaning	Symbols/units	What it is NOT	Lesson example	Phuc/common trap
Resistance R	How strongly a <i>particular component</i> opposes current — the ratio of the voltage you push across it to the current you get through it. Bigger R means less current for the same p.d.	The ratio of potential difference across a component to the current through it: $R = V/I$. A property of the <i>component</i> — it depends on the material AND the shape/size (length, area). Unit ohm.	R ; ohm Ω ($= \text{V A}^{-1}$)	NOT a property of the material alone (it depends on l and A too); NOT resistivity; NOT VI (that is power).	Lesson Q5a: a component has $V = 0.7 \text{ V}$ at $I = 40 \text{ mA}$ $\Rightarrow R = 0.7/(40 \times 10^{-3}) = 17.5 \Omega$ (44:01).	Phuc trap (verified, lesson 44:01 & 52:19): Phuc wrote $R = I \times V$ and was corrected. ALWAYS $R = V/I$, never multiply. “Never again!”

Term	Student-friendly meaning	Exact physics meaning	Symbols/units	What it is NOT	Lesson example	Phuc/common trap
Resistivity ρ	A number that says how good or bad a <i>material</i> is at conducting, regardless of the shape of the object. Low ρ = great conductor (copper); high ρ = insulator (rubber, wood).	An intrinsic property of a material defined by $R = \rho l/A$, i.e. $\rho = RA/l$. Measures how strongly the material opposes current per unit length per unit area. Depends only on material and temperature, not on shape or size.	ρ ("rho"); $\Omega \text{ m}$ (ohm-metres)	NOT resistance (that depends on shape); NOT measured in Ω/m ; does NOT change if you cut the wire shorter or thinner.	Copper $\rho \approx 1.72 \times 10^{-8} \Omega \text{ m}$; nichrome $\approx 1.1 \times 10^{-6}$ — copper is a far better conductor (lesson 24:06).	Common trap: unit is $\Omega \text{ m}$ (ohm times metre), NEVER Ω/m — the latter is dimensionally wrong since $\rho = RA/l = \Omega \cdot \text{m}^2/\text{m}$.
Ohm's Law	For a metal wire kept at the same temperature, doubling the voltage doubles the current — they stay in fixed proportion, so resistance is constant.	For a metallic conductor at constant temperature, current is <i>directly proportional</i> to potential difference: $V \propto I$, equivalently R is constant. The I - V graph is a straight line through the origin.	$V \propto I$ at constant T	NOT just " $V = IR$ " (that always holds); the LAW is the <i>proportionality at constant temperature</i> . NOT obeyed by a filament lamp.	Lesson: ohmic graph is a straight line through origin, gradient = $1/R$ (11:31, 12:05).	Common trap: say "current is <i>directly proportional</i> to p.d.", not merely "linear" — a straight line not through the origin is linear but does NOT obey Ohm's law.
Ohmic conductor	A component (like a fixed resistor or a metal wire at steady temperature) whose resistance stays the same no matter the voltage — its I - V graph is a straight line through zero.	A conductor for which $I \propto V$ at constant temperature, so $R = V/I$ is constant. Straight-line I - V characteristic through the origin.	straight I - V through origin; constant R	NOT a filament lamp or diode; NOT a curve.	Figure 10.1a: straight line through origin $\Rightarrow$ obeys Ohm's law (lesson 11:12).	Common trap: "constant resistance" $\Leftrightarrow$ straight line <i>through the origin</i> ; a curve means variable R = non-ohmic.

Term	Student-friendly meaning	Exact physics meaning	Symbols/units	What it is NOT	Lesson example	Phuc/common trap
Non-ohmic conductor	A component (like a light-bulb filament) whose resistance changes as conditions change, so the current is no longer just proportional to the voltage — the graph bends.	A conductor whose resistance is not constant, so I is not proportional to V ; the I - V graph is a curve. A filament lamp is non-ohmic because it heats up, raising ρ and hence R .	curved I - V ; variable R	NOT a straight line; NOT a fixed resistor.	Filament lamp: as current flows it heats up, $T \uparrow \Rightarrow R \uparrow$, graph curves (lesson 31:15-33:23).	Common trap: for the lamp, resistance must be read as $R = V/I$ at each point — you CANNOT use the gradient (gradient $\neq 1/R$ on a curve).
Cross-sectional area A	The area of the circular face of the wire — what you'd see looking down the end of it. Found from the diameter, not used as the diameter.	The area of cross-section perpendicular to current flow. For a circular wire $A = \pi r^2 = \pi d^2/4$, with $r = d/2$. Must be in m^2 in $R = \rho l/A$.	A ; m^2 ; $A = \pi d^2/4$	NOT the radius or diameter; NOT in mm^2 when used in SI formulae.	CP7: micrometer reads d , then $A = \pi d^2/4$ (e.g. $d = 0.191 \text{ mm} \Rightarrow A = 2.86 \times 10^{-8} \text{ m}^2$).	Common trap: $A = \pi r^2$ uses the <i>radius</i> . Substituting the <i>diameter</i> into πr^2 makes A four times too big. Use $A = \pi d^2/4$ to be safe.
Diameter d & micrometer	The width of the wire, measured with a precise screw-gauge tool because the wire is so thin a ruler can't read it accurately.	The diameter of the wire, measured with a micrometer screw gauge (resolution 0.01 mm). Convert to metres with $\times 10^{-3}$. Repeat at several positions/orientations and average.	d ; mm $\rightarrow$ m via $\times 10^{-3}$; micrometer res. 0.01 mm	NOT the radius; mm is NOT converted with $\times 10^{-6}$ (that is for micrometres μm).	Worked: $d = 5.77 \text{ mm} \Rightarrow 5.77 \times 10^{-3} \text{ m}$ (micrometer figure).	Common trap: diameter is in milli-metres (mm, $\times 10^{-3}$), NOT micro-metres (μm , $\times 10^{-6}$). Confusing them errors A by 10^6 and ρ by 10^6 .
Zero error (micrometer)	When the tool reads something other than zero with nothing in its jaws — you must subtract that offset from every reading or all your diameters are wrong.	A systematic error: the micrometer reads a non-zero value when fully closed on nothing. Note it and subtract from every diameter reading; otherwise A (and hence ρ) carries a systematic error.	systematic offset on d	NOT a random error; NOT removed by repeating/averaging.	CP7 step 2: close on nothing, record any zero error, subtract from all d (method).	Common trap: zero error is <i>systematic</i> — averaging more readings does NOT remove it; you must subtract it.

Term	Student-friendly meaning	Exact physics meaning	Symbols/units	What it is NOT	Lesson example	Phuc/common trap
Core Practical 7 (CP7)	The experiment to measure a wire's resistivity: measure R at several lengths, plot R against l , take the gradient, and multiply by the wire's area.	Determination of ρ of a wire: measure d (micrometer) $\rightarrow A$; measure $R = V/I$ at several lengths l ; plot R (y) vs l (x); gradient $= \rho/A$; so $\rho = \text{gradient} \times A$.	plot R vs l ; gradient ρ/A ; $\rho = \text{grad} \times A$	The gradient is NOT ρ itself; you do NOT plot l on the y-axis.	Constantan: gradient $7.9 \Omega \text{ m}^{-1}$, $A = 2.86 \times 10^{-8} \Rightarrow \rho = 2.3 \times 10^{-7} \Omega \text{ m}$ (SaveMyExams).	Common trap: the gradient gives ρ/A , NOT ρ . You MUST multiply by A at the end — missing this is the most common CP7 mark-scheme loss.
Why a graphical method?	Using lots of (R, l) points and a best-fit line is more reliable than one pair of readings — it averages out scatter and the line's gradient ignores any constant offset from bad contacts.	Plotting R vs l and using the gradient: (i) reduces <i>random</i> error by averaging many points; (ii) eliminates the <i>systematic</i> error of contact resistance — a non-zero intercept shows contact resistance, but the gradient is unaffected by a constant offset.	gradient method; intercept = contact resistance	NOT just to “get a nice graph”; the intercept is NOT ignored randomly — it diagnoses contact resistance.	Textbook CP: scatter = random error; non-zero intercept = contact resistance; gradient cancels it.	Common trap: a non-zero R -intercept is a <i>feature</i> , not a mistake — it reveals contact resistance, and the gradient is immune to it.
Temperature dependence (metal vs semiconductor)	Heat a metal and it conducts worse; heat a semiconductor (like a thermistor) and it conducts better — opposite behaviours.	Metals: $T \uparrow \Rightarrow$ more lattice-ion vibration $\Rightarrow$ more collisions $\Rightarrow \rho \uparrow \Rightarrow R \uparrow$. Semiconductors (NTC thermistor): $T \uparrow \Rightarrow$ more charge carriers freed $\Rightarrow \rho \downarrow \Rightarrow R \downarrow$.	metal: $\rho \uparrow$; NTC semiconductor: $\rho \downarrow$	NOT the same direction for both; the filament lamp's curve comes from the <i>metal</i> behaviour.	Lesson: filament lamp heats up, $T \uparrow \Rightarrow R \uparrow$ (32:51). This is why CP7 keeps current low.	Common trap: do not mix the directions. Metal $T \uparrow \Rightarrow R \uparrow$; NTC thermistor $T \uparrow \Rightarrow R \downarrow$.

2. Formula / case table

Quantity / case	Formula	Symbol meanings	Units	When to use	Rearrangements	Trap	Mini worked example
Resistance (Ohm's law form)	$R = \frac{V}{I}$	R = resistance; V = p.d. across component; I = current through it	R in Ω ; V in V; I in A	Finding R at any operating point (ohmic OR non-ohmic — use the actual V, I pair).	$V = IR$; $I = \frac{V}{R}$	Phuc's verified mistake: $R = V/I$ NOT $R = VI$. "Very Itchy Rash": cover R to get V/I .	Non-ohmic component: $V = 0.7$ V, $I = 40$ mA $\Rightarrow R = \frac{0.7}{40 \times 10^{-3}} = 17.5 \Omega$ (lesson Q5a).
Resistivity definition	$R = \frac{\rho l}{A}$	R = resistance; ρ = resistivity; l = length; A = cross-sectional area	R Ω ; ρ Ω m; l m; A m ²	Relating a wire's resistance to its material and geometry; the core equation of this topic.	$\rho = \frac{RA}{l}$; $l = \frac{RA}{\rho}$; $A = \frac{\rho l}{R}$	l MUST be in metres, A in m ² — convert before substituting. $R \propto l$ and $R \propto 1/A$.	Carbon chip: $\rho = 3.0 \times 10^{-5}$, $l = 0.010$ m, $A = 5.0 \times 10^{-6}$ m ² $\Rightarrow R = \frac{3.0 \times 10^{-5} \times 0.010}{5.0 \times 10^{-6}} = 0.060 \Omega$.
Find resistivity from a wire	$\rho = \frac{RA}{l}$	ρ = resistivity; R = measured resistance; A = area; l = length	Ω m	Material unknown; R, A, l all measured (single-reading version of CP7).	$R = \frac{\rho l}{A}$	Unit comes out $\frac{\Omega \cdot \text{m}^2}{\text{m}} = \Omega \text{ m}$ — quote Ω m, never Ω/m .	Nichrome: $R = 18.3 \Omega$, $l = 1.000$ m, $d = 0.2743$ mm $\Rightarrow A = \pi(0.13715 \times 10^{-3})^2 = 5.913 \times 10^{-8}$; $\rho = \frac{18.3 \times 5.913 \times 10^{-8}}{1.000} = 1.08 \times 10^{-6} \Omega \text{ m}$.
Cross-sectional area of a wire	$A = \pi r^2 = \frac{\pi d^2}{4}$	d = diameter; $r = d/2$ = radius	A in m ² (so d, r in m)	Converting a micrometer diameter reading into the area needed for $R = \rho l/A$.	$d = \sqrt{\frac{4A}{\pi}}$; $r = \sqrt{\frac{A}{\pi}}$	πr^2 uses the radius . Putting d where r goes makes A four times too large.	$d = 0.40$ mm $\Rightarrow r = 0.20 \times 10^{-3}$ m; $A = \pi(0.20 \times 10^{-3})^2 = 1.26 \times 10^{-7}$ m ² .

Quantity / case	Formula	Symbol meanings	Units	When to use	Rearrangements	Trap	Mini worked example
mm → m conversion	$d_{[m]} = d_{[mm]} \times 10^{-3}$	$d_{[mm]}$ = micrometer reading in mm	dimensionless factor 10^{-3}	Every CP7 diameter reading before computing A .	$d_{[mm]} = d_{[m]} \times 10^3$	mm uses $\times 10^{-3}$, NOT $\times 10^{-6}$ (μm). Wrong prefix errors A by 10^6 and ρ by 10^6 .	0.27 mm = 0.27×10^{-3} m = 2.7×10^{-4} m (not $\times 10^{-6}$).
CP7: straight-line form	$R = \frac{\rho}{A} l$	Compare to $y = mx$: $y = R$, $x = l$, gradient $m = \rho/A$ (line through origin)	R vs l (Ω vs m)	Analysing CP7 data: plot R (y) against l (x); the best-fit line is straight through the origin.	gradient $= \frac{\Delta R}{\Delta l} = \frac{\rho}{A}$	Plot R on the y -axis and l on the x -axis — not the other way round. Line should pass (near) the origin.	Points (0.25, 5), (0.5, 10), (0.75, 15) give gradient = $20 \Omega \text{ m}^{-1}$.
CP7: read off resistivity	$\rho = \text{gradient} \times A$	gradient = ρ/A from the R - l line; $A = \pi d^2/4$	ρ in $\Omega \text{ m}$	The final step of CP7 once you have the gradient and the area.	gradient = $\frac{\rho}{A}$; $A = \frac{\rho}{\text{gradient}}$	Most common CP7 error: gradient is ρ/A , NOT ρ . You MUST multiply by A .	Constantan: gradient $7.9 \Omega \text{ m}^{-1}$, $A = 2.86 \times 10^{-8} \text{ m}^2 \Rightarrow \rho = 7.9 \times 2.86 \times 10^{-8} = 2.3 \times 10^{-7} \Omega \text{ m}$.
Ohm's law — proportionality	$V \propto I$ (constant T) $\Leftrightarrow R$ constant	At fixed temperature, doubling V doubles I	—	Stating/identifying Ohm's law; recognising an ohmic conductor from its graph.	$R = \frac{V}{I} = \text{const}$ (gradient of I - V = $1/R$)	Say "directly proportional", not merely "linear"; the line must pass through the origin.	I - V straight line through origin with gradient $1/R$: e.g. 6.0 V at 120 mA $\Rightarrow R = 50 \Omega$.
Geometry scaling of R	$R \propto \frac{l}{A} \propto \frac{l}{d^2}$	Since $A \propto d^2$, resistance scales as l/d^2	relative (ratios)	"What happens to R if length/diameter change?" reasoning questions — no need for ρ if it stays the same.	$\frac{R_2}{R_1} = \frac{l_2/d_2^2}{l_1/d_1^2}$	Halving d quarters A (so $\times 4$ on R); changing l scales R in proportion. Track d^2 , not d .	Length L & diameter D both halved: $l \rightarrow \frac{1}{2}$ ($\times \frac{1}{2}$), $A \rightarrow \frac{1}{4}$ ($\times 4$) $\Rightarrow R \rightarrow 2R$. So $43 \Omega \rightarrow 86 \Omega$.

Quantity / case	Formula	Symbol meanings	Units	When to use	Rearrangements	Trap	Mini worked example
Temperature effect (metal)	$T \uparrow \Rightarrow \rho \uparrow \Rightarrow R \uparrow$	More lattice vibration $\Rightarrow$ more electron-ion collisions	qualitative	Explaining why a filament lamp is non-ohmic; why CP7 keeps current low.	— (direction only)	Metal goes UP with T ; do not confuse with semiconductor (down).	Filament lamp: as current heats it, ρ and R rise, so the I - V graph curves (lesson 32:51).
Temperature effect (semi-conductor)	$T \uparrow \Rightarrow \rho \downarrow \Rightarrow R \downarrow$	More charge carriers freed as T rises	qualitative	Thermistor/NTC questions; contrast with metals.	— (direction only)	NTC semiconductor goes DOWN with T — opposite to a metal.	NTC thermistor: warming it lowers ρ , lowering R and raising current.
CP7 uncertainty in area	$\frac{\delta A}{A} = 2 \frac{\delta d}{d}$	Because $A \propto d^2$, the fractional uncertainty in A is twice that in d	fractional (%)	Propagating the micrometer uncertainty into A and then ρ .	$\frac{\delta \rho}{\rho} = \frac{\delta(\text{grad})}{\text{grad}} + 2 \frac{\delta d}{d}$	The factor of 2 (from the square) is easy to forget.	$\delta d/d = 1\%$ at $d = 0.27 \text{ mm} \Rightarrow \delta A/A = 2\%$, which then adds to the gradient uncertainty for $\delta \rho/\rho$.

3. CP7 method at a glance

Stage	What you do	Key detail / mark-scheme keyword
Measure diameter	Micrometer (res. 0.01 mm) at 5–10 positions and orientations; average	“small diameter needs a high-resolution instrument”; “different positions/orientations to check non-circularity and reduce random error”
Zero-error check	Close micrometer on nothing; subtract any reading from all d	Removes a <i>systematic</i> error in A
Set up circuit	Wire taped to ruler, fixed end at $l = 0$; ammeter in series, voltmeter in parallel (or ohmmeter); flying lead sets length	Fixed end must be exactly at ruler zero (else systematic length error)
Take readings	At each l (e.g. 0.25 m steps), switch on briefly, read V and I , switch off; ≥ 3 readings, average	“Switch off between readings to prevent heating, which would change ρ ”
Compute R Plot & gradient	$R = V/I$ at each length (or read ohmmeter) R (y) vs l (x); best-fit line; gradient $= \Delta R / \Delta l = \rho / A$	$R = V/I$ — never VI Use a large triangle to cut percentage uncertainty

Stage	What you do	Key detail / mark-scheme keyword
Find ρ	$\rho = \text{gradient} \times A$, with $A = \pi \bar{d}^2/4$	MUST multiply gradient by A — gradient alone is ρ/A , not ρ

4. Big picture (last-minute recall)

- $R = V/I$ **never** $R = VI$ (Phuc's verified lesson mistake — never again).
- Resistance is a property of a *component* (depends on shape); resistivity ρ is intrinsic to the *material* (independent of shape).
- $R = \rho l/A$: longer wire $\Rightarrow$ more R ; fatter wire $\Rightarrow$ less R . Rearrange to $\rho = RA/l$.
- Unit of ρ is $\Omega \text{ m}$ (ohm-metres), **never** Ω/m .
- Area $A = \pi d^2/4 = \pi r^2$ — uses the **radius**; never put diameter into πr^2 .
- Diameter is in **mm** ($\times 10^{-3}$ to metres), NOT μm ($\times 10^{-6}$).
- CP7: plot R (y) vs l (x); gradient = ρ/A ; so $\rho = \text{gradient} \times A$ — you MUST multiply by A .
- A non-zero R -intercept on the CP7 graph = contact resistance; the gradient is immune to it (why the graphical method beats a single reading).
- Metal: $T \uparrow \Rightarrow \rho \uparrow \Rightarrow R \uparrow$. NTC semiconductor: $T \uparrow \Rightarrow \rho \downarrow \Rightarrow R \downarrow$.
- Ohmic = straight I - V line *through the origin* (constant R); say “directly proportional”, not just “linear”.